

BILKENT UNIVERSITY
MATH 241
ENGINEERING MATHEMATICS I

FINAL EXAMINATION

08-01-2014

Instructions:

- There are six questions in the exam. Solve all.
 - Duration of the exam is 150 minutes.
 - The exam is closed book, closed notes.
 - No calculators are allowed. Turn off all cell phones and tablets.
 - Please read each question carefully and justify your answers in order to get credit.
 - Do not detach pages.
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Question	Grade
1	12
2	20
3	15
4	18
5	20
6	10
Total:	95

Q1. (15 points) Let A and B be two matrices of dimensions $m \times k$ and $k \times n$ respectively, where k, m and n are positive integers. Let $\text{col}(A)$ stand for the column span of A and $\text{null}(A)$ stand for the null space of A . Show that

(a) $\text{null}(B) \subseteq \text{null}(AB)$

(b) $\text{col}(AB) \subseteq \text{col}(A)$

(c) $\text{null}(A)$ is orthogonal to $\text{col}(A^T)$.

a) $\text{null}(B)$ spans the solution of $\underline{B}\underline{x} = \underline{0}$

where B is $k \times n$ x is $n \times 1$ and 0 is $k \times 1$.

Since $\underline{B}\underline{x}$ and 0 are $k \times 1$ dimensions, we can multiply by A each side. Then.

$$\underline{A}(\underline{B}\underline{x}) = \underline{A}\underline{0} = \underline{0}_{m \times 1}$$

$$\Rightarrow (\underline{AB})\underline{x} = \underline{0}$$

This means same x vector is also solution space of $\underline{AB}$

Therefore $x \in \text{Null}(\underline{AB})$ if $x \in \text{Null}(\underline{B})$

Finally $\text{null}(\underline{B}) \subseteq \text{null}(\underline{AB})$

c) * $A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} \Rightarrow A^T = [r_1^T \ r_2^T \ \dots \ r_m^T]$

Then rows of A is columns of A^T

Null A spans x vectors such that $\underline{A}\underline{x} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} \underline{x} = \underline{0}$

$$\underline{x} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_m \underline{v}_m$$

$$\Rightarrow r_i \cdot \underline{x} = 0 \text{ for } 1 \leq i \leq m$$

$$\Rightarrow r_i = r_i^T \text{ from } (*)$$

$$(r_i \cdot x = r_i^T \cdot x) \Rightarrow \langle r_i, x \rangle = 0 \text{ for } 1 \leq i \leq m.$$

Therefore null A is orthogonal to $\text{col}(A^T)$

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Q2. (20 points) Consider the following vectors and matrix:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 5 \\ 1 \end{bmatrix}, \text{ and } A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 5 \\ -3 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} -13 & 11 & 10 \\ -19 & 16 & 15 \\ 18 & -15 & -14 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) Determine if the vectors $\{v_1, v_2, v_3\}$ are linearly independent or not.

(b) Find an orthonormal basis for $\text{Span}\{v_1, v_2, v_3\}$.

(c) Find A^{-1} by computing $\det(A)$ and $\text{adj}(A)$.

(d) Find A^{-1} by elementary row operations on the matrix $[A \ I_{3 \times 3}]$.

no
no

$$a) \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 5 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{-2R_1 + R_2} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 5 & 5 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{-\frac{R_2}{5} + R_3} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 5 & 5 \\ 0 & 1 & 1 \end{bmatrix}$$

Set $\{v_1, v_2, v_3\}$ are Linearly dependant

pivots are on 1st and 2nd columns.

whereas set $\{v_1, v_2\}$ are linearly independent

b) A basis for $\text{Span}\{v_1, v_2, v_3\} \Rightarrow \text{Span}\{v_1, v_2\}$ from part a

We need to use Gram-Schmidt Orthogonalization, but we realize v_1 and

v_2 are already orthogonal since $v_1 \cdot v_2 = -2 + 2 + 0 = 0$

$$\text{Then, } e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad e_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

orthonormal basis = $\text{Span}\{e_1, e_2\}$

$$c) \det A = \begin{vmatrix} 1 & 4 & 5 \\ 4 & 2 & 5 \\ -3 & 3 & 1 \end{vmatrix} = 1 \cdot (2 - 15) - 4(4 + 15) + 5(12 + 6)$$

$$= -13 - 76 + 90 = 1$$

$$\det A = \frac{1}{1} \text{Adj } A$$

$$A_{11} = 1 \cdot (2 - 15) = -13$$

$$A_{21} = (-1) \cdot (4 - 15) = 11$$

$$A_{12} = (-1) \cdot (4 + 15) = -19$$

$$A_{22} = 1 \cdot (1 + 15) = 16$$

$$A_{13} = 1 \cdot (12 + 6) = 18$$

$$A_{23} = (-1) \cdot (3 + 12) = -15$$

$$A_{31} = 1 \cdot (20 - 10) = 10$$

$$\text{Adj } A = \begin{bmatrix} -13 & 11 & 10 \\ -19 & 16 & 15 \\ 18 & -15 & -14 \end{bmatrix}$$

$$A_{32} = (-1) \cdot (5 - 20) = 15$$

$$A_{33} = 1 \cdot (2 - 16) = -14$$

Q3. (15 points) Let L be the line in $\mathbb{R}^3$ defined by all $\mathbf{x} \in \mathbb{R}^3$ satisfying: $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

and V be the plane in $\mathbb{R}^3$ defined by all $\mathbf{z} \in \mathbb{R}^3$ satisfying: $\begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \mathbf{z} = 0$.

(a) Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 2 & 2 & -1 \end{bmatrix}^T$ onto L .

(b) Find an orthonormal basis for V .

(c) Find a basis for the subspace $L_V = \{\mathbf{x}_V : \mathbf{x} \in L\}$ where $\mathbf{x}_V$ is the orthogonal projection of $\mathbf{x}$ onto V .

$$\begin{aligned} \text{a) } \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} x_2 + x_3 &= 0 \\ x_1 &= 0 \end{aligned} \quad L = t(0, 1, -1) \text{ line on } \mathbb{R}^3 \\ x &= \begin{bmatrix} 0, 1, -1 \end{bmatrix}^T \\ e &= \frac{x}{\|x\|} = \frac{\begin{bmatrix} 0, 1, -1 \end{bmatrix}^T}{\sqrt{2}} \end{aligned}$$

$$\text{Proj } \mathbf{y} \text{ on } \mathbf{x} = (\mathbf{y} \cdot \mathbf{e}) \mathbf{e}$$

$$= \frac{1}{\sqrt{2}} \cdot (0 + 2 + 1) \frac{\begin{bmatrix} 0, 1, -1 \end{bmatrix}^T}{\sqrt{2}}$$

$$= \frac{3}{2} \begin{bmatrix} 0, 1, -1 \end{bmatrix}^T = \begin{bmatrix} 0, \frac{3}{2}, -\frac{3}{2} \end{bmatrix}^T$$

$$\text{b) } \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \Rightarrow x - y + 2z = 0 \text{ is the plane } V \text{ on } \mathbb{R}^3$$

Let's say parameters are such that $y = t, z = s$

$$\text{then } x = y - 2z = t - 2s$$

$$\text{Then solution of } V \Rightarrow \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Then we use Gram-Schmidt orthogonalization:

$$\mathbf{e}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\sqrt{2}}$$

$$\mathbf{e}_2 = \frac{\mathbf{v}_2 - (\mathbf{e}_1 \cdot \mathbf{v}_2) \mathbf{e}_1}{\|\mathbf{v}_2 - (\mathbf{e}_1 \cdot \mathbf{v}_2) \mathbf{e}_1\|} = \frac{\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} (-2 + 0 + 0) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\| \quad \|}$$

$$= \frac{\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\| \quad \|} = \frac{\begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}}{\sqrt{3}} = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{Then orthonormal basis for } V \Rightarrow \text{span} \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Q4. Given a difference equation $x[k] = \frac{1}{4}x[k-2] + u[k]$, compute $x[k]$, $k \geq 0$ when

(a) (5 points) $u[k] = 0$ for all $k \geq 0$, $x[-1] = 4$ and $x[-2] = 0$

(b) (5 points) $u[k] = \begin{cases} 2^k & \text{for } k=0,1 \\ 0 & \text{otherwise} \end{cases}$, $x[-1] = 0$ and $x[-2] = 0$.

(c) (10 points) Find the solution $y(t)$, $t \geq 0$ for the differential equation

$\frac{d^2}{dt^2}y(t) + 2\frac{d}{dt}y(t) + 2y(t) = 2$ for $t \geq 0$, assuming the initial conditions are zero.

a) $u[k] = 0 \Rightarrow x[k] = \frac{1}{4}x[k-2]$ $x[-1] = 4$ $x[-2] = 0$

$$r^2 = \frac{1}{4} \cdot 1 \Rightarrow r_1 = \frac{1}{2} \quad r_2 = -\frac{1}{2}$$

$$x[k] = c_1 \left(\frac{1}{2}\right)^k + c_2 \left(-\frac{1}{2}\right)^k$$

$$x[-1] = 2c_1 - 2c_2 = 4$$

$$x[-2] = 4c_1 + 4c_2 = 0 \Rightarrow c_2 = -c_1$$

$$2c_1 + 2c_1 = 4c_1 = 4$$

$$c_1 = 1 \text{ and } c_2 = -1$$

$$x[k] = \left(\frac{1}{2}\right)^k - \left(-\frac{1}{2}\right)^k$$

c) $y'' + 2y' + 2y = 2$ $y(0) = 0$, $y'(0) = 0$

We use Laplace transform

$$s^2 Y(s) - s y(0) - y'(0) + 2(s Y(s) - y(0)) + 2 Y(s) = \frac{2}{s}$$

$$Y(s) (s^2 + 2s + 2) = \frac{2}{s}$$

$$Y(s) = \frac{\frac{2}{s}}{s^2 + 2s + 2} = \frac{2}{s(s^2 + 2s + 2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 2}$$

$$\frac{As^2 + 2As + 2A + Bs^2 + Cs}{s(s^2 + 2s + 2)} = \frac{2}{s} \Rightarrow A = 1$$

$$A + B = 0 \Rightarrow B = -1$$

$$2A + C = 0 \Rightarrow C = -2$$

$$Y(s) = \frac{1}{s} + \frac{-s - 2}{s^2 + 2s + 2}$$

$$Y(s) = \frac{1}{s} - \frac{s+1}{1 + (s+1)^2} = \frac{1}{1 + (s+1)^2}$$

Q5. (20 points) Consider the homogenous differential equation defined by $\dot{x} = \begin{bmatrix} -e^{2t} \\ e^{-2t} + e^{2t} \end{bmatrix}$
 $\frac{d}{dt}x(t) = Ax(t)$ where $A = \begin{bmatrix} -2 & 0 \\ 4 & 2 \end{bmatrix}$ and $x(t) \in \mathbb{R}^2$ for all $t \geq 0$.

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(a) Is this system asymptotically stable?

$$\lim_{t \rightarrow \infty} x \neq 0$$

(b) Let $A = V D V^{-1}$, where D is a diagonal matrix, determine V .

(c) Determine $x(t)$ for $t \geq 0$, when the initial condition is $x(0) = \begin{bmatrix} -1 & 2 \end{bmatrix}^T$.

(d) Find a suitable non-zero initial condition $x(0) \in \mathbb{R}^2$ so that $x(t) \rightarrow 0$ as $t \rightarrow \infty$.

$$a) A = \begin{bmatrix} -2 & 0 \\ 4 & 2 \end{bmatrix} \quad \det(A - \lambda I) = \begin{vmatrix} -2-\lambda & 0 \\ 4 & 2-\lambda \end{vmatrix} = (-2-\lambda)(2-\lambda) - 0 = 0$$

$$\lambda_1 = -2, \quad \lambda_2 = 2$$

(5)

Since there exist a nonnegative λ (i.e. $\lambda_2 = 2$), system may not be asymptotically stable and is unstable because

$$b) A = V D V^{-1} \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A e_1 = -2 e_1 \quad e_1 = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = -2 \begin{bmatrix} a \\ b \end{bmatrix} \Rightarrow$$

$$-2a = -2a$$

$$4a + 2b = -2b$$

$$4a + 4b = 0 \Rightarrow a = -b = t$$

$$A e_2 = 2 e_2 \quad e_2 = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = 2 \begin{bmatrix} c \\ d \end{bmatrix}$$

$$e_1 = t \begin{bmatrix} 1 \\ -1 \end{bmatrix} \checkmark \checkmark$$

$$-2c = 2c \Rightarrow c = 0$$

$$4c + 2d = 2d \Rightarrow d = t$$

$$\text{Then } V = [e_1 \ e_2] = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \checkmark \text{ that diagonalize } A. \quad e_2 = t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(5)

c) I use state transition matrix

$$V^{-1} = \frac{1}{1-0} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$e^{At} = V \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} V^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Q6. (10 points) Consider the homogenous differential equation defined by $\frac{d}{dt} \mathbf{x}(t) = \mathbf{A} \mathbf{x}(t)$

where $\mathbf{A}$ is a 2×2 matrix whose entries are in $\mathbb{R}$ and $\mathbf{x}(t) \in \mathbb{R}^2$ for all $t \geq 0$.

Determine $\mathbf{A}$ from the following observations:

when the initial condition is $\mathbf{x}_1(0) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ the solution is $\mathbf{x}_1(t) = \begin{bmatrix} \sin(t) - 2\cos(t) \\ \cos(t) - \sin(t) \end{bmatrix}$ for $t \geq 0$,

when the initial condition is $\mathbf{x}_2(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ the solution is $\mathbf{x}_2(t) = \begin{bmatrix} -2\sin(t) - \cos(t) \\ \cos(t) + \sin(t) \end{bmatrix}$ for $t \geq 0$.

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$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} \cos t + 2\sin t \\ -\sin t - \cos t \end{bmatrix} = \mathbf{A} \mathbf{x} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \sin t - 2\cos t \\ \cos t - \sin t \end{bmatrix}$$

$$a(\sin t - 2\cos t) + b(\cos t - \sin t) = \cos t + 2\sin t$$

$$(a-b)\sin t + (-2a+b)\cos t = \cos t + 2\sin t$$

$$\begin{aligned} a-b &= 2 \\ -2a+b &= 1 \end{aligned} \Rightarrow \begin{aligned} -a &= -3 \\ b &= -5 \end{aligned}$$

$$c(\sin t - 2\cos t) + d(\cos t - \sin t) = -\sin t - \cos t$$

$$\sin t (c-d) + \cos t (-2c+d) = -\sin t - \cos t$$

$$c-d = -1$$

$$-2c+d = -1$$

$$\Rightarrow \begin{aligned} -c &= -2 \\ c &= 2 \\ d &= 3 \end{aligned}$$

$$\mathbf{A} = \begin{bmatrix} -3 & -5 \\ 2 & 3 \end{bmatrix}$$

We will see the same result when we check second initial condition.